

End-Semester Exam

Instructions:

- The exam duration is **120 minutes**.
- Each question is worth **10 marks** each. Maximum Marks: **40**.
- Two A4-sized dual-sided, cheat sheet allowed per student.
- There will be no fractional marking. Only integer grading will be followed. Hence please show the details of your work and write down key steps of your solutions carefully.
- **DO NOT CARRY ANY DIGITAL GADGETS.** Including cell phones and calculators with you during the exam.

Questions:

- (a) Determine a, b such that the function $u = ax^3 + bxy^2$ is harmonic and find a harmonic conjugate.
- (b) Evaluate the following contour integral counterclockwise using Cauchy's Integral Formula

$$\oint_C \frac{dz}{z^2 - 1}, \quad C : |z - 1| = \pi/2$$

- (a) Write acceleration $\mathbf{a}$ in the form of $a_T \mathbf{T} + a_N \mathbf{N}$ without finding $\mathbf{T}$ and $\mathbf{N}$. For the curve:

$$\mathbf{r}(t) = (a \cos t)\mathbf{i} + (a \sin t)\mathbf{j} + bt\mathbf{k}, \text{ where } a, b \geq 0, \text{ and } a^2 + b^2 \neq 0.$$

- (b) Show that the vector field $\mathbf{F} = (y \sin z)\mathbf{i} + (x \sin z)\mathbf{j} + (xy \cos z)\mathbf{k}$ is conservative and find the potential function for it.
- (a) Show that the curl of the vector field $\mathbf{F}$ is zero:

$$\mathbf{F} = \frac{-y}{x^2 + y^2}\mathbf{i} + \frac{x}{x^2 + y^2}\mathbf{j} + z\mathbf{k}$$

- (b) Evaluate the line integral $\oint_C \mathbf{F} \cdot d\mathbf{r}$ counterclockwise around the unit circle $C : x^2 + y^2 = 1$ in the xy -plane (where $z = 0$).
 - (c) Explain why the result in (b) is not zero, even though $\text{curl } \mathbf{F} = \mathbf{0}$. Does this contradict Stokes' Theorem?
- (a) Find the area of the elliptical region cut from the plane $x + y + z = 1$ by the cylinder $x^2 + y^2 = 1$.
 - (b) Find the flux of the field $\mathbf{F} = 4x\mathbf{i} + 4y\mathbf{j} + 2\mathbf{k}$ outward (away from the z -axis) through the surface cut from the bottom of the paraboloid $z = x^2 + y^2$ by the plane $z = 1$.